2024-25 First Semester

MATH1053 Linear Algebra I

Assignment 5

Due Date: Due on 22/Oct/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student number**. Write neatly on **A4-sized** paper (*staple if necessary*) and **show your steps**.
- **Late submissions or answers without steps won't be graded.**

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1. Find the LU decomposition of the following matrix and show your steps.
$$A = \begin{pmatrix} 1 & 4 & 2 \\ -2 & 2 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

2. Evaluate the following determinant by cofactor expansion. Write your answer as a polynomial in x.
$$\begin{vmatrix} a-x & b & c \\ 1 & -x & 0 \\ 0 & 1 & -x \end{vmatrix}$$
.

3. For each of the following matrices, compute the determinant and state whether the matrix is singular or nonsingular.
a). $\begin{pmatrix} 2 & -1 & 3 \\ -1 & 2 & -2 \\ 1 & 4 & 0 \end{pmatrix}$; b). $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & -1 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 7 & 3 \end{pmatrix}$; c). $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$.

4. Let A be a nonsingular matrix. Show that $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

5. Let A and B be $n \times n$ matrices with $\det(A) = 4$ and $\det(B) = 5$. Find the value of
a) $\det(AB)$; B) $\det(kA)$, $k \neq 0$; c) $\det(2BA)$; d) $\det(A^{-1}B)$; f) $\det \begin{pmatrix} O & A \\ B & O \end{pmatrix}$

6. Suppose that A and B are $n \times n$ matrices satisfying $\det(B) = 2$ and $A = B^2$.
   (a) Are A and B singular or nonsingular? Briefly state why.
   (b) Find the value of $\det(ABA^{-1}BA^TB^TAB^{-1})$.

7. Let A be a real $m \times n$ matrix. Prove that if $A^TA = \mathbf{O}_{n\times n}$, then $A = \mathbf{O}_{m\times n}$.
(Hint: Denote $A = (a_{ij})_{m\times n}$. Write down the expression for the $(i,i)^{\text{th}}$ entry of A^TA.)